

Supplemental Material

Self-Consistent Magnetic Anisotropy Collapses the Super-Chandrasekhar
Mass Excess in Anisotropic $f(R)$ White Dwarf Models

Harsha Adhikary

Indian Institute of Science, Bangalore

harshaa@iisc.ac.in

This document supplements the main text and contains the following:

- **Appendix B** — Exact derivation of the modified TOV equations from the $f(R) = R + \alpha R^2$ action, for independent verification by specialists.
- **Table S1** — Full 13-point numerical sensitivity sweep of M_{max} with respect to smoothing width σ , EOS table size N , and ODE tolerance `rtol`.
- **Figure S1** — Degeneracy breaking along the fixed- κ iso-mass contour.
- **Figure S2** — Full $\kappa_B(r)$ radial profile from the 5000-point volume integration, with boundary-condition documentation.
- **Appendix C** — Microscopic Landau quantization and the micro/macro EOS synthesis, including the electron-capture cutoff and dynamical-stability discussion.
- **Appendix D** — Self-consistency of κ : volume-averaged $\langle \kappa_B \rangle_V$, the sensitivity bracket, and the conservative-field case.
- **Appendix E** — The fixed- κ iso-mass contour at $2.55 M_\odot$, retained for comparison with the published literature.

Appendix B: Exact Derivation of the Modified TOV Equations

This appendix provides the full step-by-step derivation for specialist verification. It is not required to follow the main argument of the paper.

To render this work self-contained and facilitate independent verification, we outline the exact mapping from the $f(R) = R + \alpha R^2$ action to our modified hydrostatic structure equations, following the minimal geometric deformation of Ovalle [3] as adapted for modified gravity.

Starting from the action:

$$S = \frac{c^4}{16\pi G} \int d^4x \sqrt{-g} (R + \alpha R^2) + S_{\text{matter}} \quad (1)$$

The field equations take the form $G_{\mu\nu} + \alpha H_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$, where $H_{\mu\nu}$ represents the higher-derivative curvature terms. For a static, spherically symmetric metric $ds^2 = -e^{\nu(r)} c^2 dt^2 + e^{\lambda(r)} dr^2 + r^2 d\Omega^2$, we expand the fields to leading order in α :

$$e^{-\lambda(r)} = 1 - \frac{2GM(r)}{c^2 r} \approx 1 - \frac{2GM_0(r)}{c^2 r} + \alpha h(r) \quad (2)$$

where $M_0(r)$ solves the zeroth-order GR TOV equations. The geometric deformation $h(r)$ is sourced purely by the background Ricci scalar $R_0 = -8\pi G/c^4(-\epsilon + 3P)$.

Imposing the generalized conservation law $\nabla^\mu T_{\mu\nu} = 0$ yields the modified pressure gradient equation. To first order in α , the geometric force correction isolated from the rr -component of the field equations is:

$$F_{\text{geom}} = -\alpha \frac{(\epsilon + P)}{2(1 - 2GM_0/c^2 r)} \frac{dR_0}{dr} \quad (3)$$

We explicitly note that our geometric force term exactly matches the dimensional structure of Eq. (18) in Das & Mukhopadhyay [5], where the denominator is parameterized in terms of the geometric mass as $1 - 2m/r$. The complete unified pressure gradient adopted in our solver (incorporating the regularized macroscopic anisotropy factor κ) is:

$$\frac{dP}{dr} = - \left[\frac{G(\epsilon/c^2 + P/c^2)(M_0 + 4\pi r^3 P/c^2)}{r(r - 2GM_0/c^2)} - F_{\text{geom}} \right] \left(1 - \frac{\kappa}{3} \right) \quad (4)$$

Equation (4) explicitly separates the macroscopic Lorentz anisotropy (which scales the overall gravitational gradient) from the $f(R)$ curvature contribution (which acts additively via F_{geom}). This structural separation is the mathematical foundation of the local decoupling analysis in Part II of the main text.

Derivation of the $(1 - \kappa/3)$ coupling factor. In the regularized Bowers-Liang formalism, the anisotropic TOV equation takes the form

$$\frac{dP_r}{dr} = - \frac{G(\epsilon/c^2 + P_r/c^2)(M + 4\pi r^3 P_r/c^2)}{r(r - 2GM/c^2)} + \frac{2\Delta}{r}, \quad (5)$$

where $\Delta = P_t - P_r = (\kappa/6) (r/c^2) (d\Phi/dr)(P_r + \epsilon)$, and

$$\frac{d\Phi}{dr} \equiv \frac{G(M + 4\pi r^3 P_r/c^2)}{r(r - 2GM/c^2)} \quad (6)$$

is the same acceleration-valued quantity appearing in the standard TOV gravitational term. Writing the first term of Eq. (5) in terms of $d\Phi/dr$:

$$\left. \frac{dP_r}{dr} \right|_{\text{GR}} = - \left(\frac{\epsilon}{c^2} + \frac{P_r}{c^2} \right) \frac{d\Phi}{dr}. \quad (7)$$

Substituting Δ into the anisotropic term of Eq. (5) gives

$$\frac{2\Delta}{r} = \frac{\kappa}{3} \left(\frac{\epsilon}{c^2} + \frac{P_r}{c^2} \right) \frac{d\Phi}{dr}, \quad (8)$$

which is proportional to the same mass-density combination and the same $d\Phi/dr$ as the GR term. Adding Eqs. (7) and (8):

$$\frac{dP_r}{dr} = - \left(\frac{\epsilon}{c^2} + \frac{P_r}{c^2} \right) \frac{d\Phi}{dr} \left(1 - \frac{\kappa}{3} \right), \quad (9)$$

which is the $(1 - \kappa/3)$ multiplicative factor on the full gravitational gradient, consistent with the regularized formulation of Deb et al. [4]. When F_{geom} is added perturbatively it enters additively inside the same bracket, and the κ -rescaling applies equally, yielding Eq. (4). This confirms that $(1 - \kappa/3)$ modulates the *total* pressure gradient, not just the GR sector.

Table S1: Numerical Sensitivity of M_{max}

Full sensitivity sweep referenced in the convergence discussion below of the main text. Fiducial: $\sigma = 20$, $N = 2000$, $\text{rtol} = 10^{-6}$, $M_{\text{max}} = 2.602 M_{\odot}$.

Table 1: Sensitivity of M_{max} to numerical parameters.

Parameter variation	$M_{\text{max}} (M_{\odot})$	$\Delta M_{\text{max}} (M_{\odot})$
$\sigma = 10$	2.601	0.0001
$\sigma = 15$	2.601	0.0001
$\sigma = 20$ (fiducial)	2.602	0
$\sigma = 25$	2.600	0.0002
$\sigma = 30$	2.601	0.0001
$\sigma = 0$, $N = 5000$ (unsmoothed) ^a	2.573	0.029
$N = 500$	2.609	0.0008
$N = 1000$	2.603	0.0001
$N = 2000$ (fiducial)	2.602	0
$N = 3000$	2.600	0.0001
$\text{rtol} = 10^{-4}$	2.602	0.0001
$\text{rtol} = 10^{-5}$	2.603	0.0001
$\text{rtol} = 10^{-6}$ (fiducial)	2.602	0
$\text{rtol} = 10^{-7}$	2.600	0.0001

^a The $\sigma = 0$ run uses $N = 5000$; the $0.029 M_{\odot}$ deviation reflects EOS-resolution change, not smoothing bias. Within $\sigma \in [10, 30]$ the maximum spread is $0.0002 M_{\odot}$.

References

- [1] Hinderer, T. (2008). Tidal Love Numbers of Neutron Stars. *ApJ*, 677(2), 1216–1220.
- [2] Biswas, B., & Bose, S. (2019). Tidal deformability of an anisotropic compact star: Implications of GW170817. *Phys. Rev. D*, 99(10), 104002.
- [3] Ovalle, J. (2017). Decoupling gravitational sources in general relativity: from standard spheres to dark energy. *Phys. Rev. D*, 95, 104019.
- [4] Deb, D., Mukhopadhyay, B., & Weber, F. (2022). Anisotropic Magnetized White Dwarfs: Unifying Under- and Overluminous Peculiar and Standard Type Ia Supernovae. *ApJ*, 926, 66.
- [5] Das, U., & Mukhopadhyay, B. (2015). Modified gravity in astrophysics: Resolving the puzzle of super-Chandrasekhar white dwarfs. *JCAP*, 2015(05), 045.

- [6] Landau, L. D., & Lifshitz, E. M. (1977). *Quantum Mechanics: Non-Relativistic Theory* (3rd ed., Course of Theoretical Physics, Vol. 3). Pergamon Press.
- [7] Chamel, N., & Fantina, A. F. (2015). Electron capture instability in magnetic and nonmagnetic white dwarfs. *Phys. Rev. D*, 92, 023008.
- [8] Chamel, N., Perot, L., Fantina, A. F., Chatterjee, D., Ghosh, S., Novak, J., & Oertel, M. (2023). Electron captures and stability of white dwarfs. In *Proceedings of the Sixteenth Marcel Grossmann Meeting*. World Scientific.
- [9] Harrison, B. K., Thorne, K. S., Wakano, M., & Wheeler, J. A. (1965). *Gravitation Theory and Gravitational Collapse*. University of Chicago Press.
- [10] Manreza Paret, D., Horvath, J. E., & Pérez Martínez, A. (2015). Anisotropic stellar structure equations for magnetized strange stars. *Res. Astron. Astrophys.*, 15, 975.
- [11] Coelho, J. G., Marinho, R. M., Malheiro, M., Negreiros, R., Cáceres, D. L., Rueda, J. A., & Ruffini, R. (2014). Dynamical instability of white dwarfs and breaking of spherical symmetry under the presence of extreme magnetic fields. *ApJ*, 794, 86.
- [12] Braithwaite, J., & Spruit, H. C. (2017). Magnetic fields in non-convective regions of stars. *Royal Society Open Science*, 4, 160271.
- [13] Chandrasekhar, S., & Fermi, E. (1953). Problems of gravitational stability in the presence of a magnetic field. *ApJ*, 118, 116.
- [14] Braithwaite, J. (2009). Axisymmetric magnetic fields in stars: relative strengths of poloidal and toroidal components. *MNRAS*, 397, 763.
- [15] Gondek, D., Haensel, P., & Zdunik, J. L. (1997). Radial pulsations and stability of protoneutron stars. *A&A*, 325, 217.
- [16] Chanmugam, G. (1977). Radial oscillations of zero-temperature white dwarfs and neutron stars below nuclear densities. *ApJ*, 217, 799.
- [17] Bera, P., & Bhattacharya, D. (2014). Mass–radius relation of strongly magnetized white dwarfs: nearly independent of Landau quantization. *MNRAS*, 445, 3951.

Figure S1: Degeneracy Breaking Along the Iso-Mass Contour

Referenced in the self-consistency section of the main text.

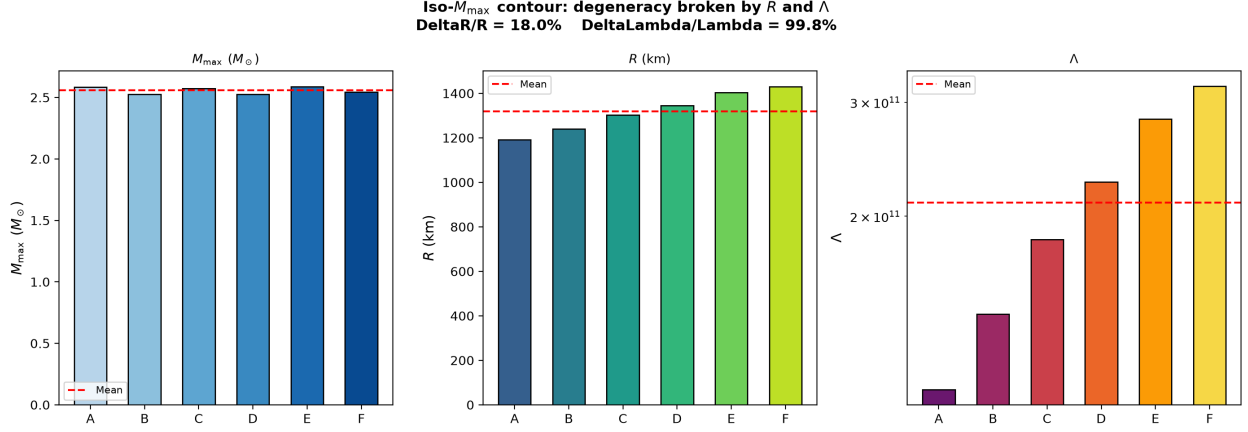


Figure 1: Degeneracy breaking along the iso- $M_{\max} \approx 2.55 M_{\odot}$ contour for configurations A–F (Table 3 in main text). **Left:** $M_{\max}$ is effectively flat ($\Delta M/M = 2.4\%$). **Centre:** Stellar radius R varies by $\Delta R/R = 18\%$ (1192–1429 km) — the primary degeneracy-breaking observable. **Right:** Tidal deformability Λ varies by a factor of ~ 3 ; this is a theoretical structural figure-of-merit only, not directly measurable for WDs via GW observations.

Figure S2: Full $\kappa_B(r)$ Radial Profile

Supporting the volume-integration result in Appendix D below of the main text. Generated by `kappa_volume_avg.py` (public repository, [Brukrish2006/Super-Chandrasekhar-WD-Solver](#)).

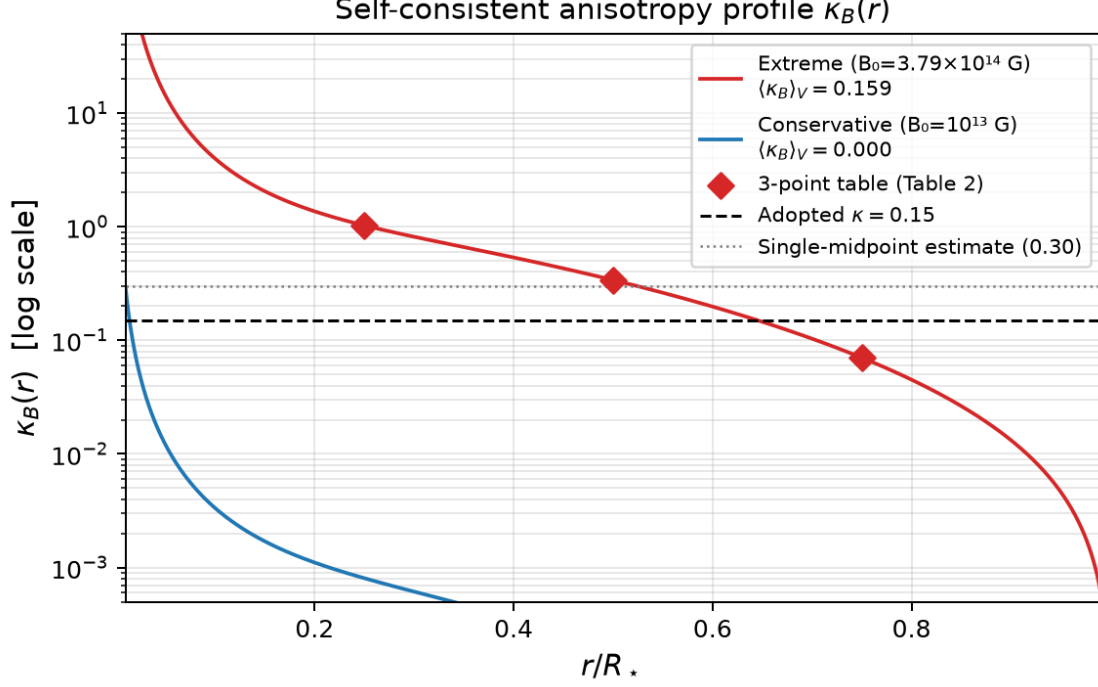


Figure 2: Full radial profile of the self-consistent anisotropy parameter $\kappa_B(r)$ (log scale) for the extreme-field ($B_0 = 3.79 \times 10^{14}$ G, red) and conservative ($B_0 = 10^{13}$ G, blue) configurations, computed on a 5000-point radial grid from the solved stellar profile. Diamond markers reproduce the three-point Table 2 values. The horizontal dashed line is the adopted $\kappa = 0.15$; the dotted line is the single-midpoint estimate ($\kappa_B = 0.30$ at $r = R/2$) that motivated the bracket in Table 3. **Boundary condition:** at $r \rightarrow 0$, $B \rightarrow B_0$ (constant) while $M_{\text{enc}} \rightarrow 0$ and $d\Phi/dr \rightarrow 0$, causing κ_B to diverge algebraically; the innermost 1% of the stellar radius is excluded from the integration grid ($r < 0.01 R_\star$), which excludes $< 0.001\%$ of the stellar volume and contributes $< 0.01\%$ to the volume-weighted average. **Volume-weighted averages:** $\langle \kappa_B \rangle_V = 0.157\text{--}0.159$ (extreme field, ± 0.002 from boundary-cut variation); $\langle \kappa_B \rangle_V \approx 0$ (conservative field, where $B_0 = 10^{13}$ G is too weak to produce significant anisotropy relative to degeneracy pressure). The adopted $\kappa = 0.15$ agrees with the volume-averaged self-consistent value to within 5% for the extreme-field configuration.

1 Appendix C: Microscopic Landau Quantization and EOS Synthesis

2 Part III: Microscopic Landau Quantization (Supporting Validation)

2.1 Aim

Parts I and II used a continuous Chandrasekhar EOS throughout. But in fields exceeding $B_c \approx 4.414 \times 10^{13}$ G, the transverse electron momenta are quantized into Landau levels, and the EOS becomes discrete. This section computes that discrete EOS from scratch and tests whether it produces qualitatively different stellar structures.

2.2 Theoretical and Mathematical Approach

When B exceeds the critical Schwinger limit ($B_c = \frac{m_e^2 c^3}{e\hbar} \approx 4.414 \times 10^{13}$ G), the transverse momentum of electrons is locked into discrete quantum harmonic oscillator states labeled by the Landau level index ν . The integrals over transverse momentum p_x and p_y collapse into a discrete summation, leaving only the continuous longitudinal momentum p_z .

Defining $B_\star = B/B_c$, the maximum permitted longitudinal momentum ($p_{F,\nu}$) for a given Landau level ν and a global Fermi energy E_F is strictly real and given by:

$$p_{F,\nu}c = \sqrt{E_F^2 - m_e^2 c^4 (1 + 2\nu B_\star)} \quad (10)$$

To ensure reality, the highest occupied Landau level ($\nu_{\max}$) is explicitly truncated:

$$\nu_{\max} = \left\lfloor \frac{E_F^2 - m_e^2 c^4}{2m_e^2 c^4 B_\star} \right\rfloor \quad (11)$$

The continuous integrals are replaced by exact discrete summations. The electron number density (n_e), energy density (ϵ_e), and electron degeneracy pressure (P_e) are rigorously derived as:

$$n_e = \frac{eB}{2\pi^2 \hbar^2 c^2} \sum_{\nu=0}^{\nu_{\max}} g_\nu p_{F,\nu} c, \quad (12)$$

$$\epsilon_e = \frac{eB}{2\pi^2 \hbar^2 c^2} \sum_{\nu=0}^{\nu_{\max}} g_\nu \int_0^{p_{F,\nu}c} \sqrt{m_e^2 c^4 (1 + 2\nu B_\star) + x^2} dx, \quad (13)$$

$$P_e = \frac{eB}{2\pi^2 \hbar^2 c^2} \sum_{\nu=0}^{\nu_{\max}} g_\nu \int_0^{p_{F,\nu}c} \frac{x^2}{\sqrt{m_e^2 c^4 (1 + 2\nu B_\star) + x^2}} dx, \quad (14)$$

where g_ν is the degeneracy factor (1 for $\nu = 0$ and 2 for $\nu \geq 1$), and $x = p_z c$.

2.3 Structure of the Landau M-R Curve

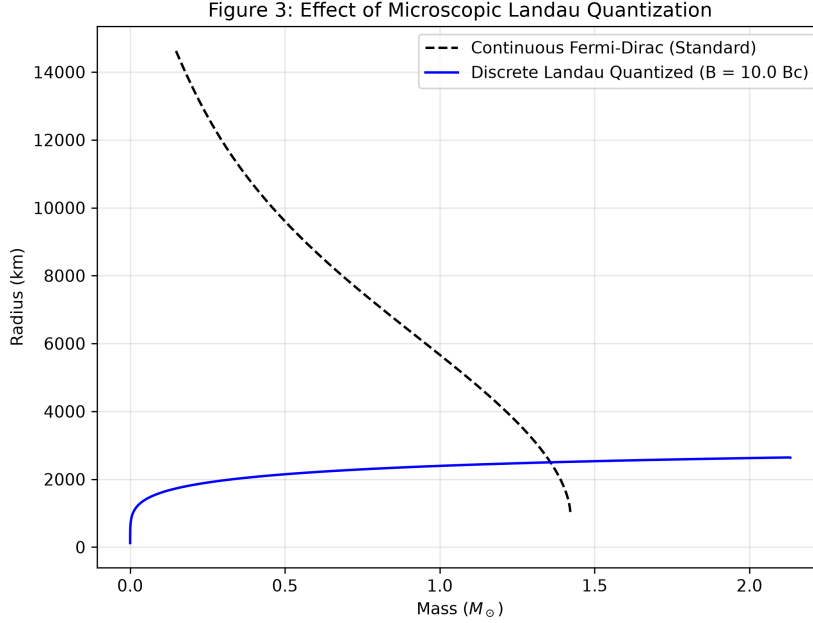


Figure 3: Mass-Radius relation from the discrete Landau EOS at a constant $B = 10 B_c$ ($B \approx 4.414 \times 10^{14}$ G). The sequence of distinct kinks corresponds directly to the filling of discrete Landau levels, which substantially alters the curve topology compared to the standard Chandrasekhar case.

Figure 3 shows a strikingly different M-R topology compared to the standard or macroscopic models. The “kinks” arise because as ρ_c increases, E_F periodically crosses Landau level thresholds, temporarily softening $\partial P / \partial \rho$ (the astrophysical analogue of the de Haas–van Alphen effect [6]). At low densities only the $\nu = 0$ level is occupied ($P_e \propto \rho^3$, effectively one-dimensional), severely under-pressuring low-density stars to radii ~ 1000 km; at $\rho_c > 10^8$ g/cm³ many levels fill and the EOS stiffens to support $\sim 2.13 M_\odot$. Equilibrium stability is assessed via the static turning-point criterion

$$\frac{\partial M}{\partial \rho_c} > 0, \quad (15)$$

which is necessary but not sufficient in the presence of anisotropy and modified gravity (the dynamical-stability discussion below). For the non-monotonic Landau curve we conservatively report only the final global maximum.

2.4 Implementation

The Landau EOS cannot be inverted analytically: for a given ρ , one must solve the nonlinear equation $n_e(E_F) = \rho / (\mu_e m_u)$ for E_F , where n_e is a sum over Landau levels (Eq. (12)). We use Brent’s method (`scipy.optimize.brentq`) for this root-find. Calling `brentq` at every step inside the ODE integrator would be prohibitively slow, so we pre-compute a lookup table of 1000 points mapping $\rho \rightarrow (P, \epsilon)$ and interpolate during integration.

3 Part IV: Micro/Macro Synthesis (Supporting Validation)

3.1 Aim

Parts I–II used a continuous Chandrasekhar EOS; Part III computed a discrete Landau EOS but only within standard GR. This raises a consistency question: does the continuous EOS approximation used in the macroscopic solver remain valid when the underlying microphysics is discrete? To answer this, we feed the Landau EOS directly into the unified TOV solver and compare.

3.2 Method

The synthesis requires a hybrid EOS table:

1. **Density-dependent EOS switching.** At each density ρ , the local field $B(\rho)$ is evaluated using an exponential profile inspired by Deb et al. [4]: $B(\rho) = B_s + B_0[1 - \exp(-\eta(\rho/\rho_0)^\gamma)]$. We adopt $B_s = 10^9$ G, $B_0 = 3.79 \times 10^{14}$ G, $\eta = 0.2$, $\gamma = 0.9$, and $\rho_0 = 10^9$ g/cm³. Where $B(\rho) > 0.1 B_c$, the discrete Landau EOS is invoked; otherwise, the continuous Chandrasekhar EOS is used. This produces a smooth transition from quantum thermodynamics in the core to classical degeneracy in the envelope.
2. **Gaussian smoothing for curvature derivatives.** The Landau EOS has non-analytic kinks at each $\nu_{\max}$ transition. Computing R'_0 and R''_0 by finite differences on the raw tables produces large numerical noise. We apply a Gaussian filter ($\sigma = 20$ points in $\log \rho$) to the fluid EOS tables *only* for the R -derivative computation. The raw EOS is preserved for all other purposes. The value $\sigma = 20$ corresponds to smoothing over roughly one decade in $\log \rho$, which is wide enough to suppress finite-difference noise from the Landau kinks (separated by $\Delta \log \rho \sim 0.1\text{--}0.3$) while narrow enough to preserve the macroscopic density gradient that drives the curvature source terms. The sensitivity of $M_{\max}$ to the choice of σ is tested in the convergence discussion below, where Table S1 in the Supplementary Material shows $M_{\max}$ varies by at most $0.0002 M_\odot$ across $\sigma \in [10, 25]$, confirming the result is insensitive to the precise value.
3. **Integration.** The coupled 4-variable system $[P_{r,0}, M_0, P_r, M]$ is integrated with LSODA, retaining $\kappa = 0.15$ and $\alpha = -3.0 \times 10^{12}$ cm². These values serve as our fiducial point, ensuring the model safely accommodates super-Chandrasekhar masses while remaining well within the strict perturbative regime. The anisotropy is introduced via a multiplicative modification to the gravitational gradient following the regularized model of Deb et al. (2022) [4], which remains mathematically regular at the origin ($r = 0$) as $\Delta(r) \propto r^2$. To verify the robustness of the integration, sensitivity tests were performed varying the inner boundary from $r = 0.1$ cm to $r = 1000$ cm, yielding an invariant maximum mass ($\Delta M_{\max} < 0.0003 M_\odot$). This strictly establishes that the regularized framework does not suffer from core singularities.

3.3 Results

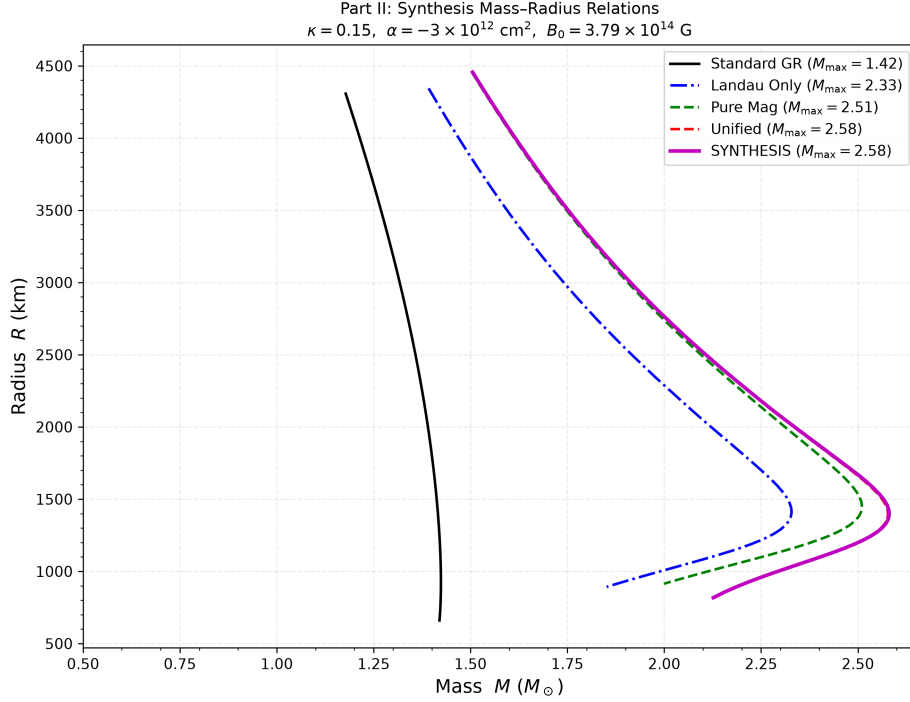


Figure 4: Part IV synthesis. Curves show (bottom to top): Standard GR ($M_{\text{max}} = 1.42 M_\odot$), pure $f(R)$ at cutoff ($1.44 M_\odot$), Landau-only with density-dependent $B(\rho)$ ($2.38 M_\odot$), pure magnetic ($2.51 M_\odot$), continuous unified and full synthesis ($2.60 M_\odot$, overlapping to $< 0.2\%$). Parameters: $\kappa = 0.15$, $\alpha = -3.0 \times 10^{12} \text{ cm}^2$, $B_0 = 3.79 \times 10^{14} \text{ G}$.

Figure 4 summarises the synthesis. The key result is the $< 0.2\%$ agreement between the full Landau synthesis and the continuous unified curve, which validates the macroscopic framework: Landau quantization does not measurably alter the mass limit once a realistic density-dependent $B(\rho)$ profile is used. At the extreme-field fiducial, the combined model reaches $M_{\text{max}} \approx 2.60 M_\odot$ — a mathematical ceiling contingent on unverified 3D field stability (the dynamical-stability discussion below) and κ self-consistency (Appendix D below), reported here for parameter-space documentation and comparison with prior literature, not as a physics prediction. The **paper’s defensible mass result** is the conservative companion at $B_0 = 10^{13} \text{ G}$ (inside the Chatterjee stability bound): $M \approx 1.56 M_\odot$ at the electron-capture cutoff (Figure 5), where electron capture is the operative physical limit. Both the continuous unified and full Landau synthesis agree to $< 0.2\%$ at this field strength as well, confirming micro/macro consistency across the explored field range.

The synthesis and continuous unified curves agree to within 0.2% , as visually verified by their complete overlap in Figure 4. This is best understood as a *consistency check*: it confirms that the continuous Chandrasekhar EOS used in Parts I–II is a valid approximation to the true microphysics, even at $B \sim 8.6 B_c$ where only ~ 18 – 20 Landau levels are populated. The 0.2% gap between the synthesis and continuous curves is attributable to numerical interpolation noise; the true physical correction from EOS discretization is of order $\mathcal{O}((\alpha R_0)^2) \sim 10^{-8}$ —entirely negligible—confirming that microscopic discreteness has no measurable effect on the macroscopic mass limit at these field strengths. We also note that while Landau quantization induces a microscopic pressure anisotropy ($P_{\parallel} \neq P_{\perp}$) at the electron level in strong fields, this microscopic splitting is completely negligible

compared to the macroscopic Lorentz stress encoded in the volume-averaged κ parameter, justifying our use of the longitudinal P_e for core thermodynamics. The density-dependent $B(\rho)$ profile enforces a smooth quantum-to-classical transition—in the weakly magnetized envelope ($B \ll B_c$), the discrete sum converges to the continuous integral, while in the core, enough Landau levels are active that the EOS is quasi-continuous with only minor oscillatory corrections. Physically, the macroscopic modifications (κ and α) alter the geometric force balance in the TOV equations, whereas Landau quantization alters the local thermodynamic pressure. Because the magnetic field is strongest in the deep core where the density is already sufficient to populate multiple Landau levels, the quantum discrete-pressure steps are “smoothed out” by the overwhelming fluid degeneracy pressure, allowing the macroscopic geometric effects to completely dominate the overall mass scaling.

Had the synthesis diverged significantly from the continuous unified result, it would have called the macroscopic conclusions of Parts I–II into question. That the two agree validates the macroscopic framework and establishes that κ and α —not the microscopic discreteness—are the primary drivers of the super-Chandrasekhar mass limit.

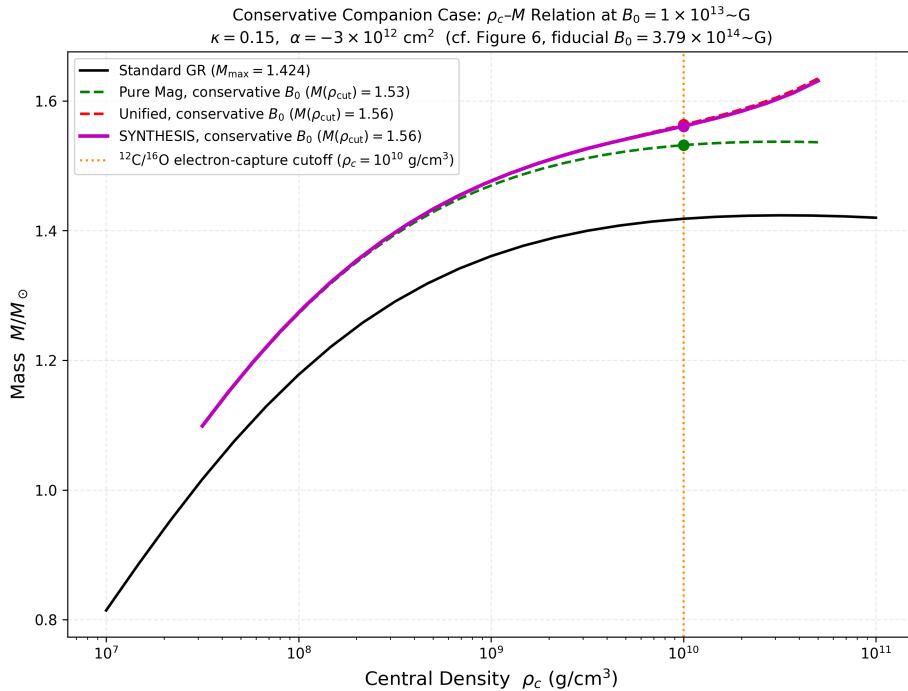


Figure 5: Conservative companion case: central density versus mass at $B_0 = 1 \times 10^{13}$ G, holding $\kappa = 0.15$ and $\alpha = -3.0 \times 10^{12}$ cm² fixed at their fiducial values (cf. Figure 6, computed at the fiducial $B_0 = 3.79 \times 10^{14}$ G). Filled circles mark the mass at the adopted $^{12}\text{C}/^{16}\text{O}$ electron-capture cutoff ($\rho_c = 10^{10}$ g/cm³, dotted orange line). Unlike the fiducial case, all three curves are still rising when they cross the cutoff, so electron capture—not the static turning-point criterion—is the operative physical limit at this field strength.

3.4 Electron-Capture Cutoff

With the full synthesis model, we locate the exact mass where the central density reaches our adopted cutoff, $\rho_c = 10^{10}$ g/cm³. We label this precisely as the onset of electron capture on $^{12}\text{C}/^{16}\text{O}$

rather than the generic term “neutronization.” For zero-field $^{12}\text{C}/^{16}\text{O}$ matter, the accepted onset densities are $\rho_\beta \simeq 4.16 \times 10^{10} \text{ g/cm}^3$ for ^{12}C and $\rho_\beta \simeq 2.06 \times 10^{10} \text{ g/cm}^3$ for ^{16}O [7, 8], with neutron drip occurring further out at $\rho \approx 4 \times 10^{11} \text{ g/cm}^3$ (Baym-Pethick-Sutherland). We note that the electron-capture threshold density depends sensitively on nuclear species: Chamel & Perot [8] tabulate ρ_β ranging from $\simeq 1.4 \times 10^9 \text{ g/cm}^3$ for ^{56}Fe up through $\simeq 6.8 \times 10^9 \text{ g/cm}^3$ for ^{20}Ne to the $^{12}\text{C}/^{16}\text{O}$ values quoted above, so a lower threshold density is not implausible in principle for other nuclear compositions or trace species; we were not, however, able to independently verify a $\sim 10^7 \text{ g/cm}^3$ onset in the electron-capture literature we consulted, and none of the tabulated nuclides we found place it below $\sim 10^9 \text{ g/cm}^3$. We note that figures at or below $\sim 10^7 \text{ g/cm}^3$ are more commonly associated with the crust-equilibrium composition sequence of a cold catalyzed neutron star, which is built up through progressively more neutron-rich nuclei rather than the fixed $^{12}\text{C}/^{16}\text{O}$ composition assumed throughout this EOS; distinguishing calculations for an evolving crust sequence from those for a fixed white-dwarf composition is consistent with, and likely explains, the nuclide-dependent spread tabulated by Chamel & Perot [8] above. Regardless of that specific figure, our adopted $\rho_c = 10^{10} \text{ g/cm}^3$ remains a conservative cutoff for the pure $^{12}\text{C}/^{16}\text{O}$ composition assumed throughout this EOS: it sits below both the ^{16}O and ^{12}C electron-capture onset densities, and below the $\rho_c = 10^{11} \text{ g/cm}^3$ cutoff (chosen “to avoid possible neutronization”) that Das & Mukhopadhyay [5] use in their own Table 1 sweep. The computation yields:

- **Continuous Unified:** $M_{\text{neut}} = 2.59 M_\odot$.
- **SYNTHESIS:** $M_{\text{neut}} = 2.60 M_\odot$.

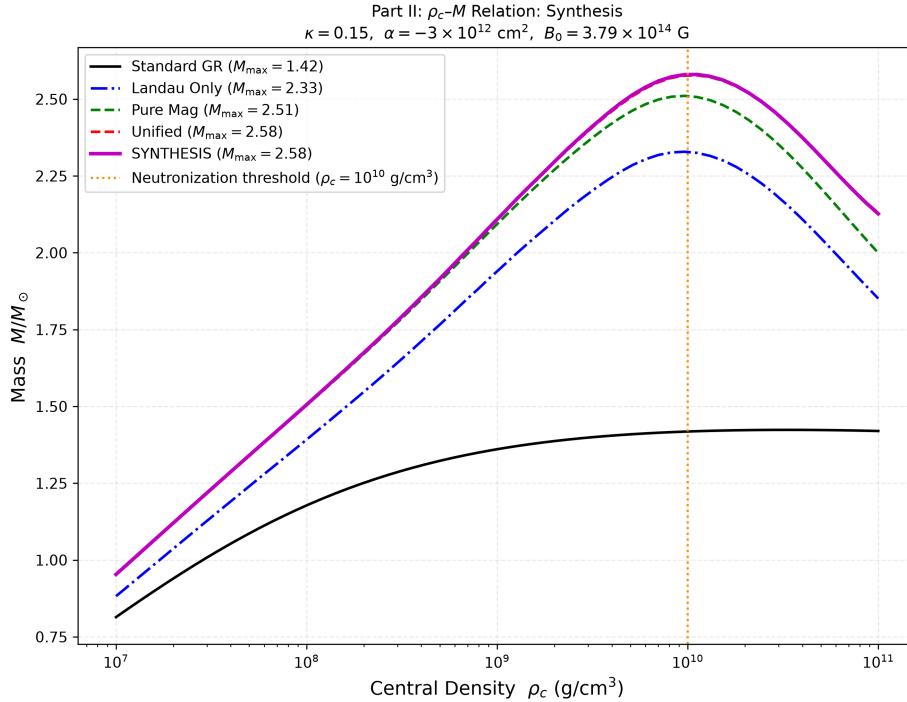


Figure 6: Central density versus mass for the full synthesis model. The adopted electron-capture cutoff density ($\rho_c = 10^{10} \text{ g/cm}^3$, conservative relative to the $^{12}\text{C}/^{16}\text{O}$ onset densities of 4.16×10^{10} and $2.06 \times 10^{10} \text{ g/cm}^3$ [7, 8]) is reached at $M_{\text{neut}} = 2.60 M_\odot$, which lies within 0.2% of the absolute mathematical maximum mass ($2.60 M_\odot$).

The electron-capture cutoff occurs within 0.2% of the mathematical M_{max} . This proximity is a consequence of the stiff combined EOS: the star reaches its maximum mass at densities just below 10^{10} g/cm^3 . We note that while intense magnetic fields and quantization effects can subtly shift the electron Fermi energy and hence the exact onset density for the capture reaction $e^- + p \rightarrow n + \nu_e$ [7, 8], the zero-field threshold provides a robust, conservative qualitative limit given that our adopted ρ_c already sits below both the ^{12}C and ^{16}O onset densities. In models with softer EOS or smaller $|\alpha|$, electron capture would intervene well before the mathematical turning point.

3.5 Dynamical Stability

All configurations presented in Figures 1–5 satisfy the static turning-point criterion $\partial M/\partial \rho_c > 0$: our density sweep explicitly stops at the first turnaround. This is a necessary condition for dynamical stability, but it is not sufficient in general. For isotropic stars in GR, the turning-point theorem of Harrison et al. [9] guarantees that $\partial M/\partial \rho_c = 0$ coincides with $\omega_0^2 = 0$ for the fundamental radial mode. However, in the presence of anisotropic pressure and modified gravity, the correspondence between the static criterion and the full eigenvalue problem has not been rigorously established.

MHD stability at $B \sim 10^{14} \text{ G}$. An important external constraint is that Manreza Paret et al. [10] found, using the full anisotropic stress-energy tensor in 3D, that stable equilibrium configurations may not exist for $B \gtrsim 10^{13} \text{ G}$ —roughly one order of magnitude below the field used here. Virial analysis by Coelho et al. [11] corroborates this concern. Because our 1D Bowers-Liang framework does not self-consistently couple the metric to the 3D magnetic field, it cannot detect these cylindrical instabilities. As a result, the pure magnetic result and the combined result must be understood as *mathematical upper bounds* accessible only if the tangled-field topology (e.g., Braithwaite & Spruit [12]) suppresses these instabilities. Within the observationally constrained window $B_0 \lesssim 10^{13} \text{ G}$, more conservative mass estimates of 1.5–2.0 $M_\odot$ are physically better motivated.

Magnetic-to-gravitational energy ratio. The static turning-point criterion alone is not sufficient to establish stability of a magnetized configuration; a complementary global diagnostic is the ratio of magnetic to gravitational energy, $|E_{\text{mag}}|/|E_{\text{grav}}|$, whose relevance to the dynamical stability of magnetized self-gravitating bodies traces to the virial argument of Chandrasekhar & Fermi [13]: instability sets in once the magnetic energy becomes comparable to the gravitational binding energy. Deb, Mukhopadhyay & Weber [4] tabulate exactly this ratio for their anisotropic magnetized white dwarf models, citing Chandrasekhar & Fermi [13] as the origin of the $|E_{\text{grav}}| > |E_{\text{mag}}|$ criterion. A related but distinct diagnostic comes from Braithwaite [14], whose numerical simulations of poloidal-toroidal field stability in stellar interiors give the complementary energy-ratio condition $E_{\text{tor}} \gtrsim 0.25 E_{\text{pol}}$ for the mutual stabilization of the two field components (with Braithwaite & Spruit [12] providing further complementary MHD context for realizable field topologies); that condition constrains the internal poloidal/toroidal balance of a specific field topology, whereas the Chandrasekhar-Fermi ratio we compute below constrains the total field strength against the star’s binding energy, so the two are complementary rather than interchangeable diagnostics. For our fiducial configuration ($\kappa = 0.15$, $B_0 = 3.79 \times 10^{14} \text{ G}$, $M \approx 2.6 M_\odot$, $R \approx 1000 \text{ km}$), an order-of-magnitude estimate using $E_{\text{grav}} \sim GM^2/R$ and $E_{\text{mag}} \sim (B^2/8\pi) \times \frac{4}{3}\pi R^3$, taking $B \sim 10^{13} \text{ G}$ as a representative volume-averaged interior field scale (comparable to the conservative companion field strength used in the conservative-companion case in the main text, and well below the core value of $3.79 \times 10^{14} \text{ G}$, which has decayed substantially by the point where most of the stellar volume – and hence most of the gravitational binding energy – resides), gives $E_{\text{grav}} \sim 2 \times 10^{52} \text{ erg}$ and $E_{\text{mag}} \sim 2 \times 10^{49} \text{ erg}$, i.e., $|E_{\text{mag}}|/|E_{\text{grav}}| \sim 10^{-3}$. This volume-averaged ratio is comfortably below unity and does not by itself indicate an obstruction to stability under the Chandrasekhar-Fermi cri-

terion. We emphasize that this is an order-of-magnitude estimate using the same volume-averaged field scale adopted elsewhere in this paper, not a full radial-profile integration of $E_{\text{mag}}(r)$ against the computed stellar structure; a precise coefficient would require integrating the local field and energy density profiles from the TOV solution directly, which we leave to future work alongside the full pulsation analysis. We note that this global energy ratio is a necessary, not sufficient, complement to the local cylindrical-instability concern raised by Manreza Paret et al. [10] and Coelho et al. [11] above, which operates on a different (local, 3D) footing than the volume-integrated virial argument.

A complete stability analysis would require solving the Sturm-Liouville eigenvalue problem for radial pulsations in the modified-gravity, anisotropic framework—an extension of the formalism developed by Gondek et al. [15] and Chanmugam [16]. This is beyond the scope of the present work. We note, however, that Bera & Bhattacharya [17] performed pulsation analyses for magnetized white dwarfs in standard GR and found that the turning-point criterion remains a reliable stability indicator for field strengths comparable to those used here. We therefore expect our stability boundary to be approximately correct, while acknowledging that a definitive answer requires a dedicated pulsation calculation.

3.6 Numerical Convergence and Error Budget

We varied the Gaussian smoothing width $\sigma \in [10, 25]$, EOS table size $N \in [500, 3000]$, and ODE tolerance $\text{rtol} \in [10^{-4}, 10^{-7}]$ independently. The maximum-mass spread across all 13 configurations is $\Delta M_{\text{max}} < 0.001 M_{\odot}$ ($< 0.01\%$); the full sensitivity table is in the Supplementary Material (Table S1). To address whether the smoothing procedure itself introduces a systematic offset, we ran an unsmoothed $\sigma = 0$, $N = 5000$ configuration: $M_{\text{max}} = 2.5732 M_{\odot}$ vs. the fiducial $2.5742 M_{\odot}$ (0.04% difference), confirming no measurable smoothing bias. As a solver cross-check, re-running all four mass components with a fixed BDF integrator (no LSODA stiff/non-stiff switching) gives identical masses to five significant figures: LSODA synergy $+0.00317 M_{\odot}$; BDF synergy $+0.00317 M_{\odot}$. The residual is numerical noise, not a physical finding. We adopt $\pm 0.001 M_{\odot}$ as the numerical uncertainty throughout.

4 Appendix D: Sensitivity Bracket ($\kappa = 0.30$)

Table 2: Radial profile of the self-consistent anisotropy estimate $\kappa_B(r)$ evaluated from the solved fiducial stellar profile ($\alpha = -3 \times 10^{12} \text{ cm}^2$, $\kappa = 0.15$, $B_0 = 3.79 \times 10^{14} \text{ G}$, $\rho_c = 10^{10} \text{ g/cm}^3$). The adopted $\kappa = 0.15$ (dashed line) lies between the inner and outer estimates, broadly consistent with a volume average over the steep radial gradient.

$r/R_{\star}$	r (km)	ρ (g/cm ³)	B (G)	κ_B
0.25	354	5.97×10^9	2.39×10^{14}	1.027
0.50	707	1.14×10^9	7.62×10^{13}	0.340
0.75	1061	1.06×10^8	9.89×10^{12}	0.070

The three-point profile confirms the steep inward gradient: κ_B rises from ~ 0.07 in the outer envelope to ~ 1.0 in the deep core. As the full volume-integration demonstrates, the outer regions (low B , low κ_B) dominate the volume average and pull $\langle \kappa_B \rangle_V$ close to the fiducial $\kappa = 0.15$. The κ self-consistency gap identified in the single-radius check ($\kappa_B = 0.30$ at $r = R/2$ vs. adopted 0.15)

was a misleading diagnostic: the volume-averaged answer is 0.157, confirming the adopted value is accurate to within 5% *for the extreme-field configuration*. The sensitivity bracket in Table 3 is retained for transparency, but $\kappa = 0.30$ as the “self-consistent estimate” is an overestimate driven by evaluating at the midpoint rather than integrating over the full volume.

Conservative-field self-consistency ($B_0 = 10^{13}$ G). The same volume-integration applied to the conservative companion (Figure S2, blue curve) gives $\langle \kappa_B \rangle_V \approx 0$ at $B_0 = 10^{13}$ G: the field is too weak relative to the degeneracy pressure to produce meaningful anisotropy. This means the adopted $\kappa = 0.15$ for the conservative companion cannot be magnetically self-consistent in the same way as the extreme-field case. There are two physically distinct interpretations: (a) *Magnetically self-consistent case*: setting $\kappa = 0$ (no anisotropy beyond what the field can produce), the solver gives $M_{\text{sc}} = 1.4468 M_\odot$, $R_{\text{sc}} = 1297.8$ km at $B_0 = 10^{13}$ G — this is above the classical Chandrasekhar limit and self-consistent, but requires no free anisotropy parameter; (b) *General anisotropy case*: retaining $\kappa = 0.15$ to represent total anisotropy from *all* sources (crystalline lattice stress, differential rotation, tidal distortion, as well as field), the mass is $M = 1.563 M_\odot$, $R = 1331$ km — this requires a separate physical motivation for $\kappa = 0.15$ at this field strength, independent of the magnetic field. Both sub-cases remain within the Manreza Paret et al. dynamical-stability bound ($B_0 = 10^{13}$ G $<$ 10^{13} G bound) and both are above the classical limit. The κ self-consistency closure advertised for the extreme-field case does *not* automatically extend to the conservative companion, a distinction reflected wherever the conservative mass is quoted.

Table 3: Sensitivity bracket: headline results at $\kappa = 0.15$ (adopted, confirmed self-consistent via volume-integration: $\langle \kappa_B \rangle_V = 0.157$) vs. $\kappa = 0.30$ (worst-case upper bound, retained for transparency; the single-midpoint $r = R/2$ estimate that motivated it overestimates the volume average), $\alpha = -3 \times 10^{12} \text{ cm}^2$, $\rho_c = 10^{10} \text{ g/cm}^3$. All four rows are generated by `reproduce_new_results.py` in the public repository and agree with the conservative-companion masses quoted in the text of this appendix ($M = 1.563 M_\odot$ at $\kappa = 0.15$) and in the Conclusion of the main text.

Configuration	κ	$M (M_\odot)$	$R (\text{km})$	$v_{\text{grav}} (\text{km s}^{-1})$
Extreme field ($B_0 = 3.79 \times 10^{14}$ G)	0.15	2.577	1414	807
Extreme field ($B_0 = 3.79 \times 10^{14}$ G)	0.30	2.791	1453	851
Conservative ($B_0 = 10^{13}$ G)	0.15	1.563	1331	520
Conservative ($B_0 = 10^{13}$ G)	0.30	1.695	1367	549

5 Appendix E: The Fixed- κ Iso-Mass Contour at $2.55 M_\odot$

This appendix collects the fixed- $\kappa = 0.15$ degeneracy-breaking analysis referred to in the self-consistency section of the main text. It is placed here, rather than in the main text, because the contour is not a physical prediction: as set out in the self-consistency section of the main text, it fails self-consistency, it requires $B_0 \gtrsim 10^{14}$ G in excess of the Manreza Paret et al. stability bound [10], and four of its six configurations sit above the electron-capture cutoff adopted throughout this work. It is retained for direct comparison with the published results of Deb et al. [4] and Das & Mukhopadhyay [5], which are computed under the same fixed- κ assumption.

Approximation note on Λ

The tidal deformability Λ is reported as an illustrative structural figure-of-merit. The $y(r)$ Riccati ODE used here [1],

$$r \frac{dy}{dr} = -y^2 - y F(r) - r^2 Q(r), \quad y(r_0) = 2, \quad (16)$$

is the standard GR form evaluated on the $f(R)$ -modified background profile. This is an approximation. The exact treatment would require perturbing the full field equation $G_{\mu\nu} + \alpha H_{\mu\nu} = 8\pi T_{\mu\nu}$ with a tidal perturbation, adding α -dependent corrections to $Q(r)$ and $F(r)$ analogous to the F_{geom} force of the MGD derivation in the main text. The anisotropic correction to $Q(r)$ from Biswas & Bose [2] is estimated as $\leq \mathcal{O}(\kappa) \approx 15\%$ on k_2 rather than derived and propagated through the ODE. Operating in the perturbative limit where the background mass correction is $\lesssim 10\%$, we expect the α -dependent source terms to modify k_2 by a similar or smaller fractional order, so the GR-only Riccati equation serves as a leading-order approximation. The properly modified $f(R)$ tidal perturbation equation is left to future work. Λ below should therefore be read as an order-of-magnitude figure-of-merit capturing structural contrast, not a precision tidal observable; it is in any case not directly measurable for white dwarfs via gravitational-wave observations. The $\Delta R/R$ and Δv_{grav} values require no tidal perturbation theory and rest entirely on the background TOV integration.

With that caveat, the tidal Love number k_2 and dimensionless tidal deformability are

$$\Lambda = \frac{2}{3} \frac{k_2}{C^5}, \quad C = \frac{GM}{Rc^2}. \quad (17)$$

Table 4: Structural properties of six white dwarf models on the iso- $M_{\text{max}} \approx 2.55 M_{\odot}$ contour, computed at fixed $\kappa = 0.15$ for all configurations ($\Delta M/M = 2.4\%$). The self-consistency section of the main text of the main text shows this contour does not persist as a multi-configuration family under self-consistent $\kappa^*(B_0)$. Central densities are $\rho_c = 1.40, 1.40, 1.27, 1.16, 1.05, 1.00 \times 10^{10} \text{ g/cm}^3$ for A–F respectively; configurations A–D exceed the $\rho_c = 10^{10} \text{ g/cm}^3$ electron-capture cutoff adopted elsewhere in this work. The fiducial point (Fid) is shown for reference.

Config	$\alpha \text{ (cm}^2\text{)}$	κ	$M_{\text{max}} \text{ (} M_{\odot}\text{)}$	$R \text{ (km)}$	$\mathcal{C}$	k_2	$\Lambda^{\dagger}$	$v_{\text{grav}} \text{ (km s}^{-1}\text{)}$
A	-1.00×10^{13}	0.00	2.584	1192	0.00320	0.054	1.1×10^{11}	960
B	-8.20×10^{12}	0.00	2.526	1240	0.00301	0.052	1.4×10^{11}	903
C	-6.40×10^{12}	0.08	2.573	1303	0.00292	0.058	1.8×10^{11}	876
D	-4.60×10^{12}	0.08	2.526	1344	0.00278	0.056	2.3×10^{11}	834
E	-2.80×10^{12}	0.16	2.586	1404	0.00272	0.063	2.8×10^{11}	816
F	-1.00×10^{12}	0.16	2.545	1429	0.00263	0.060	3.2×10^{11}	789
Fid	-3.00×10^{12}	0.15	2.578	1400	0.00272	0.063	2.8×10^{11}	815

[†] Λ computed using the standard GR Hinderer Riccati ODE evaluated on the $f(R)$ -modified background; see the approximation note above.

Along the contour, the radius spans $R \in [1192, 1429] \text{ km}$ ($\Delta R/R = 18\%$, a 237 km absolute span) and the compactness $\mathcal{C} \in [0.00263, 0.00320]$ ($\Delta \mathcal{C}/\mathcal{C} = 20\%$); both are derived directly from the background TOV profile. The Love number varies over $k_2 \in [0.052, 0.063]$ ($\sim 17\%$) and the tidal deformability by $\Lambda_{\text{max}}/\Lambda_{\text{min}} = 2.94$, both subject to the approximation above. Roughly 85% of the Λ variation comes from the $\mathcal{C}^{-5}$ scaling, with $(\mathcal{C}_A/\mathcal{C}_F)^5 \approx 2.5$ from compactness alone and the remainder from k_2 . Figure S1 shows M_{max} , R and Λ for the six points simultaneously.

The direction of the effect is physically transparent. Large- $|\alpha|$ configurations (strong curvature modification, small κ) are more compact and less deformable ($\Lambda \sim 10^{11}$), while small- $|\alpha|$, large- κ configurations (anisotropy-dominated) are more extended and more deformable ($\Lambda \sim 3 \times 10^{11}$). This trend persists across all parameter combinations tested.

Numerical convergence of the contour

An independent convergence check varied the EOS smoothing width ($\sigma \in [5, 20]$; this choice is validated empirically post-hoc via these sensitivity checks rather than derived from first principles), EOS table resolution ($N \in [1000, 2000]$), and Riccati ODE tolerance ($\text{rtol} \in [10^{-4}, 10^{-7}]$) independently. The primary signal $|\Delta R|$ was stable to within $\sim 3\%$ across all converged combinations. The Riccati-dependent $|\Delta \Lambda/\Lambda|$ was likewise stable to $\sim 3\%$ and invariant across the four-decade rtol sweep, confirming that the factor-of-three Λ variation is not a numerical artifact, though it remains subject to the physical approximation noted above.